ANTS: Adversarial Agentic Task Synthesis

Team project · Meta Global Hackathon · Work in progress

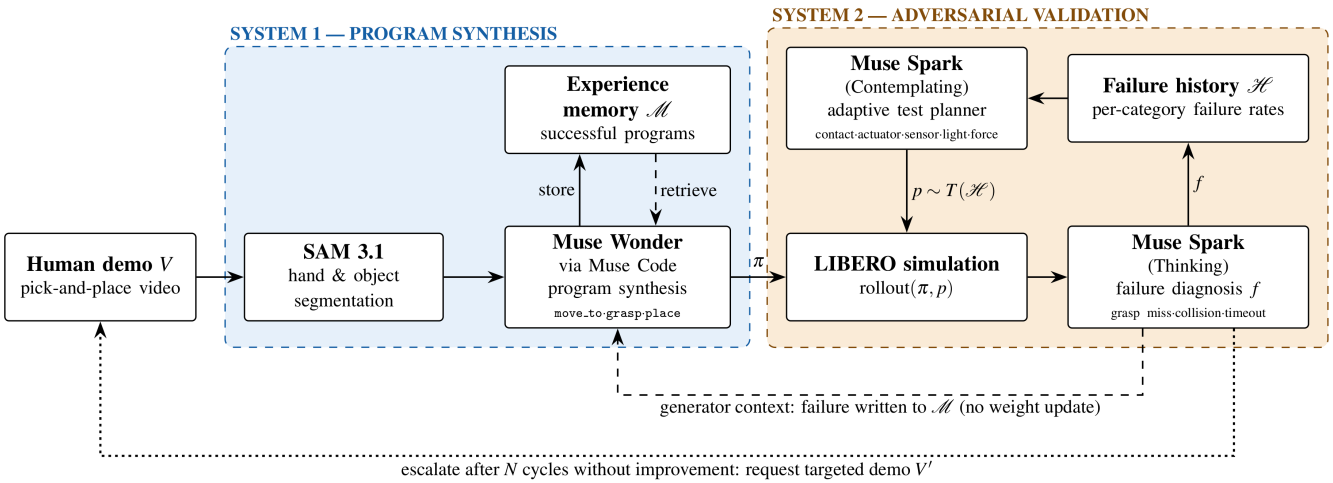
SAM 3.1 | Muse Wonder | Muse Spark | LIBERO

Overview

WORK IN PROGRESS | META GLOBAL HACKATHON

Idea: teach a robot a task from **one demo**, then let an adversary break the skill. A frozen model writes the policy. Nothing is trained. The skill improves through **context** only.

ANTS is a project for the Meta Global Hackathon. The code is on GitHub.



ANTS architecture: System 1 writes the skill, System 2 tests it, and the loop feeds each failure back.

Problem

One robot, one task, one policy. Each new task or robot needs new data and fine-tuning. Data is the bottleneck.

The Closed Loop

1. **Write (System 1):** SAM 3.1 tracks the hand, the object and the gripper in one demo video. Muse Wonder turns the result into a policy.
2. **Break (System 2):** Muse Spark plans tests. A LIBERO simulation runs them. Muse Spark explains each failure.
3. **Fix:** each failure becomes a constraint. The constraint goes into the context of the generator, never into the weights. The generator writes a new policy.
4. **Gate:** a new policy is accepted only if it is no worse on every case seen so far and better on at least one.
5. **Stop or ask:** stop when every category is robust. If the generator stalls, ask a human for one targeted demo.

Failure Cases, Not Random Tests

The adversary does **not** pick test cases at random. It looks for **failure cases**. It spends the test budget on the categories that are most likely to break the policy. Each test stays inside a bounded severity.

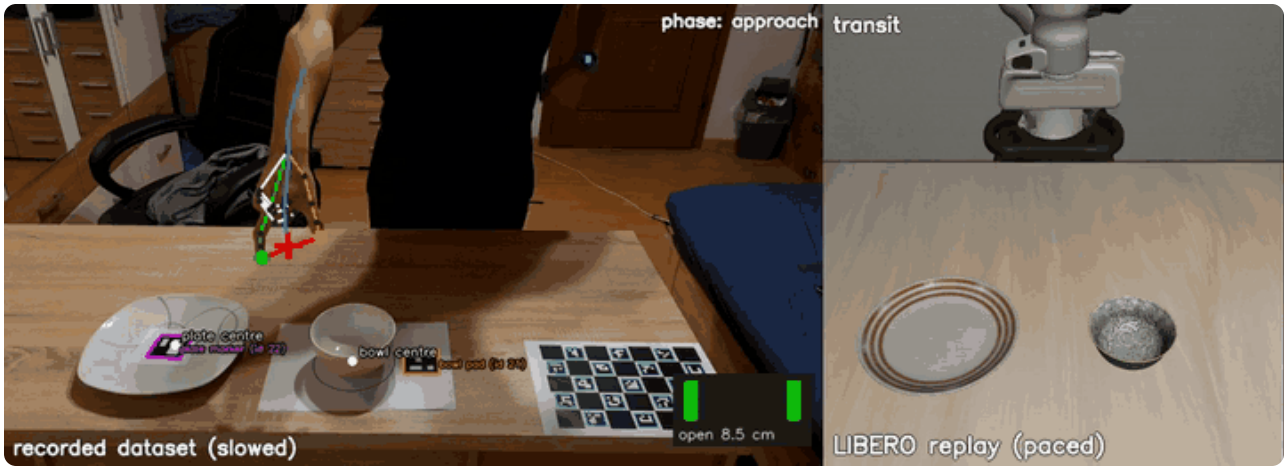
The gate keeps the result safe. A policy that fixes one failure case cannot break a case that passed before. Robustness never goes down.

Software Stack

Block	System	Tool
Hand and object segmentation	1	SAM 3.1
Policy synthesis, with experience memory	1	Muse Wonder (via Muse Code)
Test planner, uses the failure history	2	Muse Spark
Rollout of the policy under a perturbation	2	LIBERO simulation
Failure diagnosis (grasp miss, collision, timeout)	2	Muse Spark

Link to hurobo

hurobo turns a phone video of a human hand into robot demos in LIBERO. ANTS also starts from a human video and runs in LIBERO. The clip below is from hurobo: the recorded human demo (left) beside its robot replay (right).



Human demo and robot replay, synchronised (from hurobo).

Results

Results coming. All numbers will come from logged runs on a held-out test set.

Alfredo Wijaya Hardjoprawiro — M.Sc. Artificial Intelligence, TH Ingolstadt

alh7017@thi.de alfredowijaya@gmail.com GitHub LinkedIn alfredowh.github.io